

Final Sensor Characterization Report

LM35DZ Waterproof Probe and DS18B20

EE2120 Mini Project Distributed Smart Temperature Monitoring and Sensor Fusion
Network

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Characterization range 29.375 °C to 60.5625 °C

Executive Summary

The three experiments establish a measured model for the two temperature channels used in the EE2120 node. The LM35DZ analog channel is highly linear after calibration, but it carries visible ADC noise. The DS18B20 was quantized but completely steady during the noise run. In the hot-water step test, the DS18B20 waterproof probe reached its 63.2 percent target faster than the waterproof LM35DZ assembly. This measured ordering is the opposite of the generic expectation for a bare analog sensor and must guide the final fusion design.

Measured quantity	LM35DZ channel	DS18B20 channel
Calibration model	$T = 0.081567 \text{ ADC} + 14.5187 \text{ }^{\circ}\text{C}$	Reference in Experiment 1
Sensitivity or configured resolution	0.08157 $^{\circ}\text{C}$ per ADC count	0.0625 $^{\circ}\text{C}$ at 12 bit; 0.5 $^{\circ}\text{C}$ at 9 bit
Steady-state sample standard deviation	0.5425 $^{\circ}\text{C}$	0.0000 $^{\circ}\text{C}$ observed
Thermal-shock time constant	13.96 s	8.04 s
Primary design implication	Filter before fusion	Use variance floor despite flat run

The calibration fit gives $R^2 = 0.999828$ and $\text{MAE} = 0.0810 \text{ }^{\circ}\text{C}$ across 5,582 recorded pairs. The noise test gives sample variances of $0.294331 \text{ }^{\circ}\text{C}^2$ for the LM35DZ channel and $0.000000 \text{ }^{\circ}\text{C}^2$ for the DS18B20 run. The latter is an observed data variance, not a claim of perfect accuracy: all digital readings remained at one 9-bit code of 26.5 $^{\circ}\text{C}$.

1 Objective and Scope

This report converts the raw laboratory logs from three characterization experiments into parameters required for calibration, sensor comparison, and later sensor fusion. The work covers static calibration of the LM35DZ channel against the DS18B20, steady-state noise at room temperature, and a thermal step response in hot water. All numerical results were recalculated directly from the supplied CSV files.

- Derive the LM35DZ calibration equation, sensitivity, coefficient of determination, and mean absolute error.
- Measure the sample variance and standard deviation of both channels without analog oversampling.
- Estimate each probe assembly time constant from the 63.2 percent response threshold.
- Translate the measured behavior into design constraints for filtering and sensor fusion.

2 Sensor System and Test Configuration

The sensing node used an ESP32 with two independent temperature channels. The LM35DZ analog output was acquired through GPIO34 on ADC1 with 11 dB attenuation. The DS18B20 used a 1-Wire connection on GPIO4 with a 4.7 k Ω pull-up to 3.3 V. The two waterproof probe assemblies were placed together so that both experienced the same nominal bath transition.

Element	Connection or setting	Purpose
LM35DZ waterproof assembly	5 V supply; output to ESP32 GPIO34 ADC1 CH6	Analog temperature channel
DS18B20 waterproof probe	3.3 V; DQ to GPIO4; 4.7 k Ω pull-up	Digital reference and comparison channel
ESP32 acquisition	12-bit ADC hardware; 11 dB attenuation on GPIO34	Raw analog conversion

2.1 Acquisition Settings by Experiment

Experiment	LM35DZ acquisition	DS18B20 setting	Nominal interval
1 Temperature sweep	Mean of 100 ADC readings per record	12 bit	About 1.5 s
2 Steady-state noise	One raw ADC reading per record	9 bit nonblocking	0.2 s
3 Thermal shock	One raw ADC reading per record	9 bit nonblocking	0.2 s

The change in DS18B20 resolution matters when interpreting the results. Experiment 1 resolves 0.0625 °C per code at 12 bit. Experiments 2 and 3 resolve 0.5 °C per code at 9 bit, which explains the staircase transient and the flat 26.5 °C noise record.

3 Analysis Method and Data Quality

3.1 Calculation Methods

Experiment 1 used ordinary least squares with averaged ADC count as the independent variable and DS18B20 temperature as the reference variable. Sensitivity is the fitted slope per ADC count. R^2 measures the fraction of reference-temperature variation described by the linear model, and MAE is the mean absolute calibration residual.

$$T = m \cdot ADC + c$$

Experiment 2 used the sample variance with denominator n minus 1, matching the VAR.S definition in the laboratory guide. Experiment 3 used the first-order 63.2 percent threshold for a rising step. Crossing times were linearly interpolated between adjacent samples, and the time constant was measured from the selected shock-start point.

$$T_{63.2} = T_i + 0.632 \cdot (T_f - T_i) \quad \text{and} \quad \tau = t_{63.2} - t_{start}$$

3.2 Data Quality Controls

Dataset	Rows	Observed issue	Treatment and effect
Experiment 1	5,582	35 timestamp reversals and 93 repeated timestamps	Time was excluded from regression; all valid ADC-temperature pairs were retained
Experiment 2	934	One out-of-order timestamp entry	Elapsed time reconstructed from the 0.2 s sampling interval; statistics unchanged
Experiment 3	376	Initial 92 rows repeat one stale timestamp and later timestamps restart	Elapsed time reconstructed from row order; time differences after the shock remain valid

Timestamp repair affects only the horizontal axis of the visualizations. It does not change the Experiment 1 regression, the Experiment 2 distribution statistics, or the Experiment 3 difference between shock time and threshold-crossing time.

4 Experiment 1 Gradual Temperature Sweep

4.1 Method

The bound probes were exposed to a slowly changing hot-water condition. Each record averaged 100 LM35DZ ADC samples, while the DS18B20 operated at 12-bit resolution. The dataset spans 181.17 to 566.24 averaged ADC counts and 29.375 to 60.5625 °C. A linear fit was calculated from all 5,582 measurement pairs.

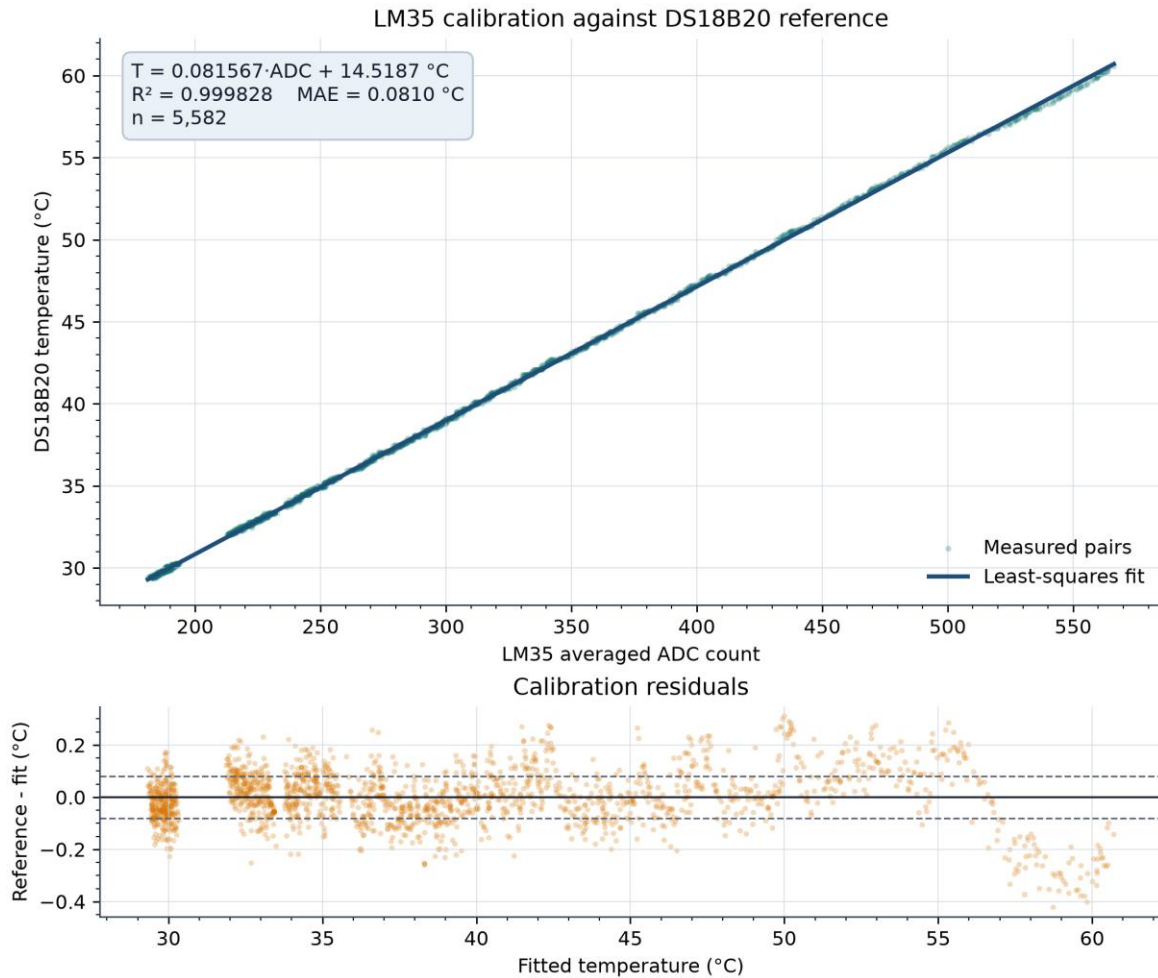


Figure 1 LM35DZ calibration scatter, least-squares fit, and residual structure across the measured range

4.2 Results

$$T_{LM35} = 0.081567 \cdot \text{ADC} + 14.5187 \text{ } ^\circ\text{C}$$

Metric	Verified result	Interpretation
Sensitivity	0.081567 °C per ADC count	Empirical system sensitivity after ESP32 conversion
Intercept	14.5187 °C	Corrects the installed channel offset over the fitted range
R ²	0.999828	Near-linear mapping across the test range
MAE	0.0810 °C	Mean absolute difference from the DS18B20 reference
RMSE	0.1057 °C	Residual magnitude with greater weight on larger errors

The fit is excellent over the measured range, although the residual plot shows structured deviations near the upper end. The equation should therefore be used as an empirical calibration for this ESP32 channel and probe assembly, not as a universal LM35DZ transfer function.

5 Experiment 2 Steady State Noise

5.1 Method

Both probes remained stationary at room temperature while the ESP32 logged one raw ADC sample and one nonblocking DS18B20 reading every 200 ms. No analog oversampling was applied. The 934 records represent 186.8 s of reconstructed elapsed time.

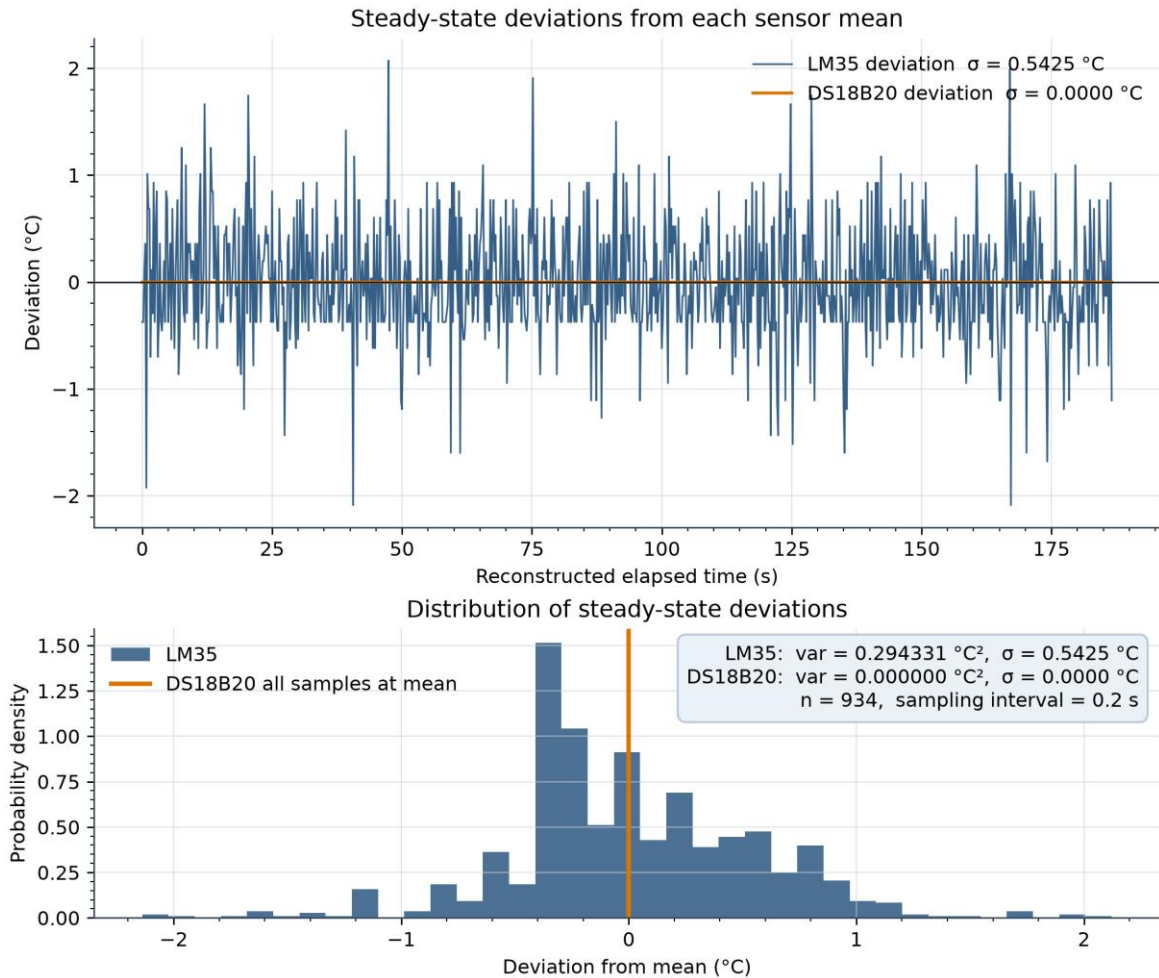


Figure 2 Time-domain deviation and distribution of steady-state readings

5.2 Results

Metric	LM35DZ channel	DS18B20 channel
Samples	934	934
Mean temperature	26.6395 °C	26.5000 °C
Sample variance	0.294331 °C ²	0.000000 °C ²
Sample standard deviation	0.5425 °C	0.0000 °C

Metric	LM35DZ channel	DS18B20 channel
Observed range	24.5515 to 28.7114 °C	26.5000 °C only

The LM35DZ channel shows substantial high-frequency variation from the analog path and ESP32 ADC. The DS18B20 record is flat because its temperature stayed within one 0.5 °C code during the run. A fusion algorithm must not interpret the zero sample variance as zero uncertainty; inverse-variance weighting would otherwise become singular and overtrust the digital channel.

6 Experiment 3 Thermal Shock Response

6.1 Method

After a room-temperature baseline, the bound probes were moved into hot water. The analysis uses a 5 s pre-transition baseline and the mean of the final 40 records as the available end-level estimate. The plotted shock start is 37.6 s on the reconstructed record, shown as 0 s in Figure 3. Threshold crossings were linearly interpolated.

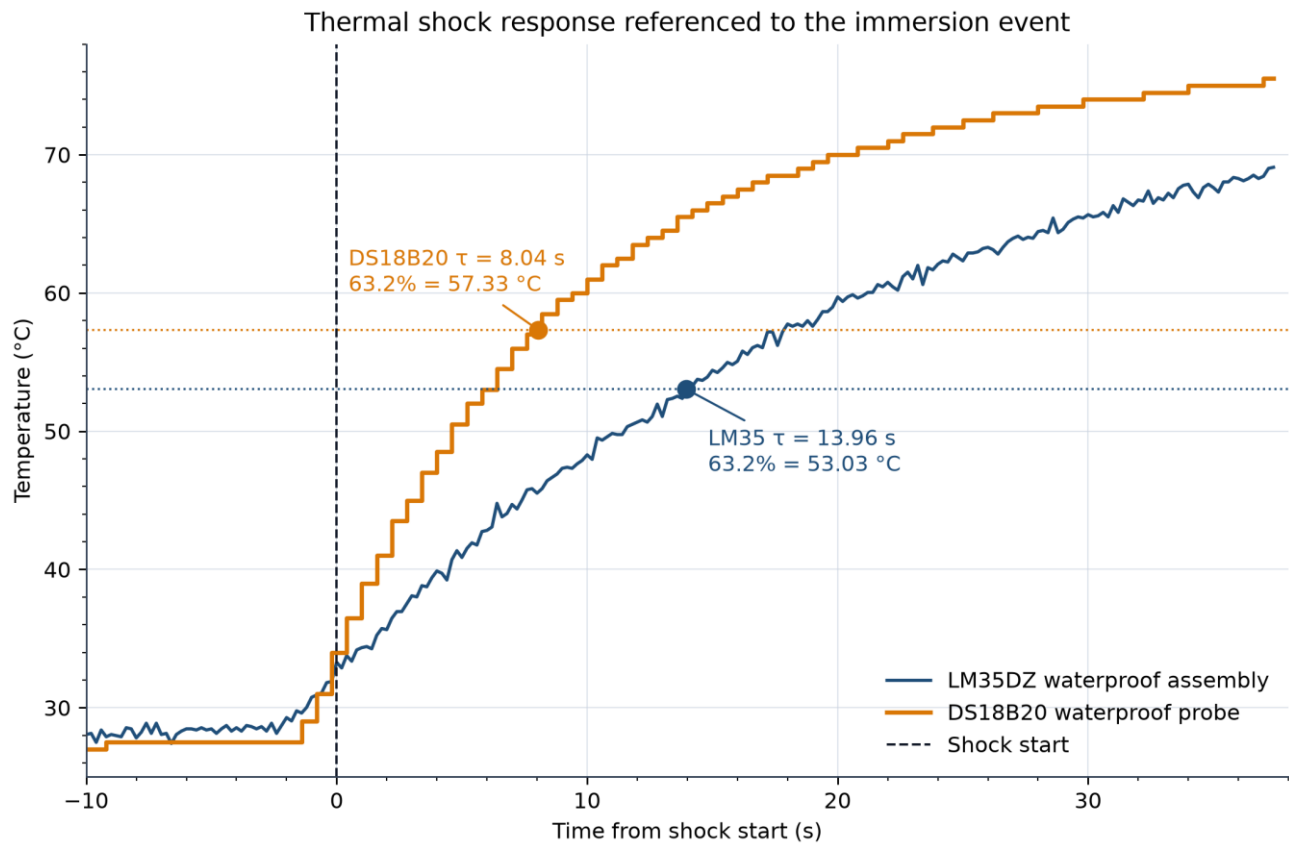


Figure 3 Hot-water step response with 63.2 percent targets and interpolated time constants

6.2 Results

Parameter	LM35DZ assembly	DS18B20 probe
Initial baseline	28.7864 °C	27.6800 °C
Final-window mean	67.1458 °C	74.5875 °C

Parameter	LM35DZ assembly	DS18B20 probe
63.2 percent target	53.0296 °C	57.3255 °C
Interpolated time constant	13.96 s	8.04 s

The DS18B20 reached its target about 5.92 s earlier. Its measured time constant was approximately 0.58 times the LM35DZ assembly time constant. In this hardware, the waterproof LM35DZ housing and encapsulation dominated the transient response, so the installed analog probe was slower even though a bare LM35 die is commonly described as fast.

7 Consolidated Sensor Comparison

Characteristic	LM35DZ waterproof channel	DS18B20 waterproof channel	Design consequence
Static calibration	Empirical linear model required	Used as Experiment 1 reference	Apply calibration before comparison or fusion
Resolution in Experiment 1	0.08157 °C per ADC count	0.0625 °C at 12 bit	Both resolve sub-degree changes during sweep
Resolution in Experiments 2 and 3	0.08157 °C per ADC count before noise	0.5 °C at 9 bit	Digital flatness partly reflects quantization
Calibration tracking error	MAE 0.0810 °C	Reference channel	Static agreement is strong over 29.4 to 60.6 °C
Steady-state noise	σ 0.5425 °C	σ 0.0000 °C observed	Filter analog data and impose a digital variance floor
Thermal response	τ 13.96 s	τ 8.04 s	Do not assume the analog probe is the faster channel
Interface	Analog ADC on GPIO34	1-Wire on GPIO4	Maintain independent channels for diagnostics

8 Sensor Fusion and Design Implications

The measured data support a state-dependent fusion strategy rather than one fixed pair of weights. Under steady conditions, the DS18B20 should receive greater weight because the recorded digital channel did not vary, while the LM35DZ needs smoothing. For example, a 10-sample moving average would reduce the standard error of independent LM35DZ samples from 0.5425 °C to about 0.1716 °C, although correlation in real ADC noise may reduce that improvement.

A zero covariance must not be assigned to the DS18B20. A practical estimator should use a positive variance floor that covers quantization, stated sensor accuracy, bath gradients, and calibration uncertainty. At 9 bit, the 0.5 °C code width alone means that a flat sequence can hide temperature movement within a quantization interval.

During transients, the measured DS18B20 probe led the LM35DZ assembly. Dynamic weighting should therefore follow the observed probe-level time constants, or each channel should be modeled with its own lag. Retaining separate raw channels remains important: the roughly 7.4 °C difference between the final-window

means in Experiment 3 can flag incomplete equilibration, mechanical contact differences, or a local thermal gradient.

- Apply the verified LM35DZ calibration equation before filtering or fusion.
- Use a moving average or low-pass filter on the analog channel, with delay chosen against the 13.96 s probe response.
- Set a nonzero DS18B20 measurement covariance even when a short record has zero sample variance.
- Preserve both sensor values and a common timestamp in telemetry so that drift, lag, and faults remain observable.

9 Limitations

- The DS18B20 served as the calibration reference but was not checked against a traceable laboratory thermometer.
- The calibration is validated only from 29.375 to 60.5625 °C and should not be extrapolated without additional tests.
- The noise result comes from one room-temperature run; the DS18B20 remained on one 9-bit output code throughout.
- Timestamp anomalies required reconstructed elapsed time. This preserves time differences at the fixed sampling interval but weakens absolute event timing.
- The hot-water bath reached the mid-70 °C range, above the originally planned 55 to 60 °C condition.
- The LM35DZ channel had not fully reached the DS18B20 final temperature when logging ended, so its 13.96 s value is an operational estimate based on the recorded final window rather than a fully settled asymptote.
- Only one thermal-shock trial was available. Repeated runs are needed for confidence intervals and run-to-run repeatability.

10 Final Conclusions

The LM35DZ channel can be calibrated accurately against the DS18B20 over the tested range using $T = 0.081567 \text{ ADC} + 14.5187 \text{ }^{\circ}\text{C}$. The fit explains 99.9828 percent of the reference variation and produces an MAE of 0.0810 °C. Static calibration is therefore not the limiting factor in this installation; unfiltered analog noise and probe dynamics are more important.

At steady state, the LM35DZ standard deviation was 0.5425 °C, while every DS18B20 sample remained at 26.5 °C. The digital channel is the better steady-state anchor in this dataset, but its zero observed variance is a quantization outcome and cannot be used as zero uncertainty.

The hot-water response test measured time constants of about 13.96 s for the waterproof LM35DZ assembly and 8.04 s for the DS18B20 probe. The final system should therefore be tuned to the installed probes rather than generic sensor expectations. The recommended design keeps both channels independent, calibrates and filters the analog data, assigns a positive uncertainty floor to the digital data, and adapts fusion weights to steady and transient operation.

Appendix A Verified Data Records

Record	Primary contents	Use in this report
Experiment1_Temperature_Sweep.csv	Timestamp, averaged ADC, 12-bit DS18B20 temperature	Calibration, sensitivity, R^2 , MAE, residuals

Record	Primary contents	Use in this report
Experiment2_Steady_State_Noise.csv	Timestamp, raw ADC, calibrated LM35DZ, 9-bit DS18B20	Means, sample variance, standard deviation, noise plots
Experiment3_Thermal_Shock.csv	Timestamp, raw ADC, calibrated LM35DZ, 9-bit DS18B20	63.2 percent targets and time constants
Experiment firmware and logger files	ESP32 acquisition settings and Python serial logging	Sampling and resolution verification

Appendix B Project References

1. EE2120 Sensor Characterization Laboratory Guide LM35 versus DS18B20 Performance Analysis.
2. EE2120 Mini Project Phase 1 Technical Literature Review and Sensor Technology Selection.
3. EE2120 Mini Project Phase 2 System Architecture and Design Report.
4. Texas Instruments LM35 Precision Centigrade Temperature Sensors datasheet.
5. Maxim Integrated DS18B20 Programmable Resolution 1 Wire Digital Thermometer datasheet.